

SIMATS ENGINEERING
ASSESSMENT TOOL 3
MCQ Test

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COURSE NAME: Computer Networks
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 20%

QUESTIONS:20

Total Marks:20

1. Which TCP mechanism reduces the congestion window by half upon detecting packet loss via duplicate ACKs?

- a) Slow Start
- b) Fast Retransmit
- ☒ c) Fast Recovery
- d) Tahoe Reset

Correct Answer: c) Fast Recovery

Explanation:

Fast Recovery is triggered after three duplicate ACKs. It reduces the congestion window (CWND) to half of its current value (and sets Ssthresh to half) instead of restarting from 1 MSS. This improves performance by avoiding Slow Start.

2. A TCP sender has $CWND = 8$ MSS and $Ssthresh = 16$. After one RTT with no loss, CWND becomes:

- a) 9 MSS
- b) 16 MSS

c) 12 MSS

d) 8 MSS

Correct Answer: b) 16 MSS

Explanation:

Since $CWND (8 \text{ MSS}) < SSTHRESH (16 \text{ MSS})$, TCP is in the Slow Start phase. During Slow Start, CWND doubles every RTT.

After one RTT:

$$CWND = 8 \times 2 = 16 \text{ MSS}$$

3. Which field in the TCP header is used to implement flow control at the receiver side?

a) Sequence number

b) Acknowledgement number

☒ c) Window size

d) Urgent pointer

Correct Answer: c) Window size

Explanation:

The Window Size field tells the sender how much data the receiver can accept without overflowing its buffer. This is TCP's flow control mechanism.

4. Nagle's algorithm causes increased latency primarily because it:

a) Disables ACK piggybacking

☒ b) Buffers small segments until ACK arrives

c) Reduces CWND on every packet

d) Increases SYN backlog

Correct Answer: b) Buffers small segments until ACK arrives

Explanation:

Nagle's Algorithm combines small packets into larger ones by waiting for an ACK before sending additional small segments. This reduces network overhead but increases latency.

5. In a high-latency satellite link (600 ms RTT), TCP throughput is primarily limited by:

- a) MTU size
- ☒ b) CWND growth rate relative to RTT
- c) DNS resolution time
- d) UDP competition

Correct Answer: b) CWND growth rate relative to RTT

Explanation:

On long RTT links, TCP takes more time to increase its congestion window. Therefore, throughput is mainly limited by the slow growth of CWND.

6. UDP is preferred over TCP for DNS queries primarily because:

- a) UDP supports fragmentation
- ☒ b) DNS responses are small and connection overhead is unnecessary
- c) TCP cannot use port 53
- d) UDP has better error correction

Correct Answer: b) DNS responses are small and connection overhead is unnecessary

Explanation:

Most DNS queries involve small request-response messages. UDP avoids the overhead of TCP's three-way handshake, making DNS lookups faster.

7. A UDP-based streaming application experiences 15% packet loss. The best application-layer mitigation is:

- a) Increase UDP buffer size
- b) Switch to TCP
- ☒ c) Implement Forward Error Correction (FEC)
- d) Reduce frame rate to 1 fps

Correct Answer: c) Implement Forward Error Correction (FEC)

Explanation:

FEC adds redundant information that allows lost packets to be reconstructed without retransmission, making it suitable for real-time streaming.

8. What is the maximum UDP payload size without IP-level fragmentation on a standard Ethernet link (MTU = 1500 bytes)?

- a) 1480 bytes
- ☒ b) 1472 bytes
- c) 1460 bytes
- d) 1500 bytes

Correct Answer: b) 1472 bytes

Explanation:

- Ethernet MTU = 1500 bytes
- IP header = 20 bytes
- UDP header = 8 bytes

Maximum UDP payload = $1500 - 20 - 8 = 1472$ bytes

9. QUIC protocol uses UDP as its transport. The key advantage over TCP-based TLS is:

- a) Eliminating IP header overhead
- ☒ b) 0-RTT or 1-RTT connection setup
- c) Removing encryption
- d) Native multicast support

Correct Answer: b) 0-RTT or 1-RTT connection setup

Explanation:

QUIC combines transport and encryption, allowing faster connection establishment using 0-RTT or 1-RTT, reducing latency.

10. HTTP/1.1 persistent connections reduce latency by:

- a) Enabling UDP fallback
- ☒ b) Reusing the same TCP connection for multiple requests
- c) Compressing HTTP headers
- d) Sending requests without waiting for DNS

Correct Answer: b) Reusing the same TCP connection for multiple requests

Explanation:

Persistent connections eliminate repeated TCP handshakes by reusing one TCP connection for multiple HTTP requests.

11. Head-of-line blocking in HTTP/1.1 occurs when:

- a) The DNS server is slow
- ☒ b) A response blocks subsequent responses on the same connection
- c) TCP CWND drops to zero
- d) The server rejects pipelined requests

Correct Answer: b) A response blocks subsequent responses on the same connection

Explanation:

In HTTP/1.1, requests are processed sequentially. A delayed response prevents later responses from being delivered.

12. HTTP/2 multiplexing solves HOL blocking at the HTTP layer but not at the:

- a) Application layer
- b) DNS layer
- ☒ c) TCP layer
- d) IP layer

Correct Answer: c) TCP layer

Explanation:

HTTP/2 multiplexes streams, but since all streams share one TCP connection, packet loss still causes TCP-level Head-of-Line blocking.

13. Which HTTP status code indicates that a resource has permanently moved and browsers should update bookmarks?

- ☒ a) 301
- b) 302
- c) 304
- d) 307

Correct Answer: a) 301

Explanation:

301 Moved Permanently tells browsers and search engines to update stored links to the new URL.

14. In HTTP caching, the Cache-Control: no-cache directive means:

- a) Never cache this resource
- ☒ b) Validate with the server before using a cached copy
- c) Cache for 0 seconds
- d) Disable all caching headers

Correct Answer: b) Validate with the server before using a cached copy

Explanation:

no-cache allows storage of the resource but requires revalidation with the server before reuse.

15. Which DNS record type maps a domain name to an IPv6 address?

- a) A
- b) CNAME
- ☒ c) AAAA
- d) MX

Correct Answer: c) AAAA

Explanation:

The AAAA record stores the IPv6 address of a domain, while the A record stores the IPv4 address.

16. A DNS resolver switches from UDP to TCP when:

- a) The query type is MX
- ☒ b) The response is truncated (TC bit set)
- c) TTL exceeds 3600 s
- d) The authoritative server is unreachable

Correct Answer: b) The response is truncated (TC bit set)

Explanation:

If a DNS response is too large for UDP, the server sets the TC (Truncated) bit, prompting the resolver to retry using TCP.

17. Which attack exploits the UDP-based DNS protocol by sending forged responses with a matching transaction ID?

- a) ARP poisoning
- ☒ b) DNS cache poisoning

- c) SYN flooding
- d) BGP hijacking

Correct Answer: b) DNS cache poisoning

Explanation:

Attackers send forged DNS replies with matching transaction IDs to trick DNS resolvers into storing incorrect IP addresses.

18. Setting a very low TTL (e.g., 30 s) for a DNS record primarily increases:

- a) Cache hit ratio
- ☒ b) Recursive resolver query load
- c) DNSSEC signature validity
- d) Zone transfer frequency

Correct Answer: b) Recursive resolver query load

Explanation:

A low TTL causes cached entries to expire quickly, resulting in more frequent DNS queries to authoritative servers.

19. A user browses to a new website. The correct sequence of protocol interactions for the first byte of the page is:

- a) HTTP → TCP → DNS
- ☒ b) DNS → TCP handshake → HTTP GET
- c) TCP → DNS → HTTP
- d) DNS → HTTP → TCP

Correct Answer: b) DNS → TCP handshake → HTTP GET

Explanation:

The browser first resolves the domain using DNS, then establishes a TCP connection, and finally sends an HTTP GET request.

20. Under 20% packet loss, which protocol combination gives the best performance for a real-time multiplayer game?

a) TCP for game state, HTTP for assets

☒ b) UDP with application-layer sequencing for game state, HTTP/3 for assets

c) HTTP/2 for all traffic

d) TCP with Nagle enabled for all traffic

Correct Answer: b) UDP with application-layer sequencing for game state, HTTP/3 for assets

Explanation:

Real-time games prioritize low latency over perfect reliability. UDP with application-layer sequencing minimizes delay, while HTTP/3 (QUIC over UDP) efficiently delivers game assets even under packet loss.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions